electronics-toys_wave1_ORDERING

Wave 1 electronics toys: ordering guide (what to buy, where, in what order)

> Companion to `MANUFACTURING_MANUAL.md`. Main caveat for this project: AP1 board order-ready; MP3 module and 4-digit display module are bought-in. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: `manuals/ORDERING_GUIDE.md`.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/ap1/1_fabrication/ap1_audio_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/ap1/2_assembly/ap1_audio_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/ap1/2_assembly/ap1_audio_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/ap1_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J2	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964
J1	BATT 3xAA	JST B2B-PH-K-S(LF)(SN)	C131337
BZ1	Buzzer	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869
J8	MP3 module left row (VCC RX TX DACR DACL SPK1 GND SPK2)	hanxia HX PM2.54-1x8P ZC-Y	C22438158
J9	MP3 module right row (BUSY .. GND IO1)	hanxia HX PM2.54-1x8P ZC-Y	C22438158
J10	Speaker 3 W 4-8 ohm	JST B2B-PH-K-S(LF)(SN)	C131337
J11	Display module (VCC SCLK RCLK DIO GND)	JST S5B-PH-K-S(LF)(SN)	C157923
J13	RGB LED (common anode) / lamp	JST B4B-PH-K-S(LF)(SN)	C131334
J14	Projector motor	JST B2B-PH-K-S(LF)(SN)	C131337
TP1, TP2, TP4	TP		

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US6.61*for1board* * *, * * *US*6.08 each for 10 boards**, **US$5.03 each for 100 boards** (about INR 582 / 535 / 443 at 88 INR/USD). The per-line break-up is in `costing/ap1_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
U1	1	ATmega328P-AU	ATMEGA328P-AU	C14877	2.3308	2.331
C1,C2,C3,C21,C25,C24,C22	7	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075	0.052
R7,R47	2	10k	0603 10k 1%	C25804	0.0024	0.005
J2	1	ISP	HCTL PZ254-2-03-Z-8.5 (ZHOURI C5116479 had only 340 in stock)	C2894964	0.0766	0.077
J1	1	BATT 3xAA	JST B2B-PH-K-S(LF)(SN)	C131337	0.0359	0.036
SW1	1	Power	C&K PCM12SMTR	C221841	1.2754	1.275
Q2	1	AO3401A	AO3401A	C15127	0.0969	0.097
C4	1	10u	0805 10uF X5R	C15850	0.0441	0.044

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
BZ1	1	Buzzer	YUEXIN KC1206 16 ohm passive magnetic 12x9.5 mm (2048 Hz)	C781869	0.1513	0.151
Q1	1	MMBT3904	MMBT3904	C20526	0.0118	0.012
D2	1	1N4148WS	1N4148WS	C2128	0.0120	0.012
R6,R5,R8,R10,R11,R60,R61,R62,R63,R64,R65,R66	12	1k	0603 1k 1%	C21190	0.0017	0.020
D1	1	Green	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	0.0105	0.011
D4	1	Amber	Hubei KENTO KT-0603Y yellow (Everlight 19-213 was out of stock)	C2287	0.0105	0.011
J8	1	MP3 module left row (VCC RX TX DACR DACL SPK1 GND SPK2)	hanxia HX PM2.54-1x8P ZC-Y	C22438158	0.1359	0.136
J9	1	MP3 module right row (BUSY .. GND IO1)	hanxia HX PM2.54-1x8P ZC-Y	C22438158	0.1359	0.136
C20	1	100u/35V	Lelon VZH101M1VTR-0607	C176683	0.0881	0.088
J10	1	Speaker 3 W 4-8 ohm	JST B2B-PH-K-S(LF) (SN)	C131337	0.0359	0.036
J11	1	Display module (VCC SCLK RCLK DIO GND)	JST S5B-PH-K-S(LF) (SN)	C157923	0.0777	0.078
U5	1	74HC595	74HC595D	C5947	0.1079	0.108
D10,D11,D12,D13,D14,D15,D16	7	Red	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	0.0077	0.054
SW2	1	Button 1	Omron B3U-1000P	C231329	0.1894	0.189
SW3	1	Button 2	Omron B3U-1000P	C231329	0.1894	0.189
SW4	1	Button 3	Omron B3U-1000P	C231329	0.1894	0.189
SW5	1	Button 4	Omron B3U-1000P	C231329	0.1894	0.189
SW6	1	Button 5	Omron B3U-1000P	C231329	0.1894	0.189
SW7	1	Button 6	Omron B3U-1000P	C231329	0.1894	0.189
SW8	1	Button 7	Omron B3U-1000P	C231329	0.1894	0.189

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
R40	1	1k	UNI-ROYAL 0603WAF1001T5E	C21190	0.0017	0.002
R41	1	2.2k	UNI-ROYAL 0603WAF2201T5E	C4190	0.0018	0.002
R42	1	3.9k	UNI-ROYAL 0603WAF3901T5E	C23018	0.0021	0.002
R43	1	6.8k	UNI-ROYAL 0603WAF6801T5E	C23212	0.0024	0.002
R44	1	12k	UNI-ROYAL 0603WAF1202T5E	C22790	0.0031	0.003
R45	1	22k	UNI-ROYAL 0603WAF2202T5E	C31850	0.0042	0.004
R46	1	47k	UNI-ROYAL 0603WAF4702T5E	C25819	0.0020	0.002
Q9,Q10,Q11,Q12	4	AO3400A	AO3400A	C20917	0.0899	0.360
R50,R51,R52,R53	4	100	UNI-ROYAL 0603WAF1000T5E	C22775	0.0024	0.010
R54,R55,R56,R57,R58,R59	6	100k	0603 100k 1%	C25803	0.0030	0.018
J13	1	RGB LED (common anode) / lamp	JST B4B-PH-K-S(LF) (SN)	C131334	0.0458	0.046
J14	1	Projector motor	JST B2B-PH-K-S(LF) (SN)	C131337	0.0359	0.036
D8	1	SS14	SS14	C84343	0.0252	0.025
TP1,TP2,TP4	3	TP				

2b. Enclosure: printed parts and fasteners (generated from the board)

- Print `4_enclosure/` base and lid (outer about 143 x 107 x 50 mm) in PETG or ABS, 3 perimeters, 25 % infill; the lid closes with **4 x M2.5 x 10 mm self-tapping screws** into the corner columns; the board sits on its standoffs with 4 x M2.5 x 6 mm self-tapping screws.
- Battery: **Keystone 2465 (3 x AA, wire leads)** in the floor bay (60.0 x 50.3 x 17.8 mm); print `*_enclosure_battery_door.stl`; the door closes with 2 x M2.5 x 6 mm self-tapping (tool needed to open). Route the holder lead through the hole in the bay ceiling to the battery connector.
- Print `*_enclosure_caps.stl` (light pipe, plunger, slider): plunger for SW2; plunger for SW3; plunger for SW4; plunger for SW5; plunger for SW6; plunger for SW7; plunger for SW8; light pipe for D1; light pipe for D4; slider for SW1. Light pipes in clear PETG or resin; fit plungers and pipes from inside before closing the lid.
- shared platform box: display window for the bought 4-digit module (fix the module to the lid behind tinted acrylic with M3 screws/standoffs; lead to J11), seven button plungers, speaker grille to add for the chosen speaker; 3 x AA bay with a screwed door
- Print one set first and test-fit the board, the holder and every cap; the dimensions are first-pass.

3. Bought-in and mechanical items

Bought-in items (not on the JLCPCB assembly)

Item	Qty per toy	Used in	Where (India)	Notes
DFPlayer Mini MP3 module (or clone with the same pinout)	1	P1 rhymes, P4 cricket (P3 optional)	Robu.in, Robocraze, Amazon.in (about Rs 120-180)	Check the command codes on the real module (see firmware README).
microSD card 1-8 GB, FAT32	1	with the MP3 module	any shop	Files `/mp3/0001.mp3`... as in the prototype guide.
Speaker 3 W, 4-8 ohm, 40-57 mm, with 2-pin JST PH lead	1	P1, P4	Robu.in, Amazon.in	
4-digit 0.36 in LED display module with two 74HC595 (pins VCC SCLK RCLK DIO GND)	1	P4 cricket	Robu.in, Robocraze, Amazon.in, AliExpress (about Rs 60-120)	Common-anode version expected; if digits look inverted or ghosted, change `DISP_SEG_ACTIVE_LOW` / `DISP_DIGIT_ACTIVE_HIGH` in `firmware/src_avr/main_ap1.cpp` and rebuild.
JST PH 2.0 mm 5-pin to Dupont female lead, 15-20 cm	1	P4 cricket	Robu.in, Amazon.in, AliExpress	Plugs J11 to the display module header. Check the pin order end to end before power-up.
Keystone 2465 (or equivalent) 3 x AA holder with lead and 2-pin JST PH plug	1	all	element14 India, Amazon.in	
RGB common-anode LED (or LED strip segment, 5 V) with 4-pin JST PH lead	1	P3 lamp	Robu.in	
Small 3-6 V DC motor for the projector disc, 2-pin JST PH lead	1	P3 lamp (optional)	Robu.in	
Tinted acrylic window, M3 x 8 screws + 4 mm standoffs for the display module	1 set	P4 cricket	local laser cutter, hardware shop	
AA cells	3	all	any shop	

4. Confidence (read before you order)

- **Order confidence: Medium-High**: limited stock on: BZ1 Buzzer (stock 140).
- **Works-as-intended confidence: Low-Medium**: AP1 board added (ATmega328P, MP3 module socket, bought-in 4-digit 74HC595 display module on a 5-way lead, 7-LED ball lane on a 74HC595, 7-button resistor ladder, lamp and motor drivers); all three apps now build for the board (about 6 kB each) and pass 76 host checks. The MP3 command codes are from memory and must be checked on the real module; the ladder thresholds are calculated, not measured.
- **First thing to do:** Bench-test one board per product: ladder readings, MP3 commands, display module polarity (one compile flag each).
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.